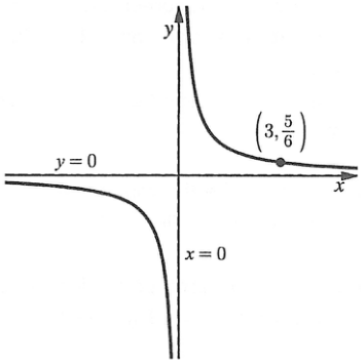
Graph Sketching: Rational Functions

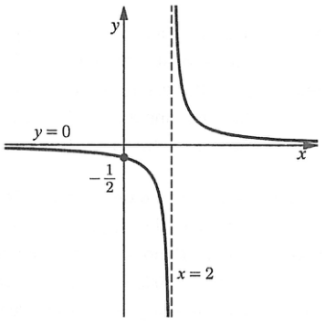
Questions 1–10

Question 1. Find the equation of the following graph in the form $y = \frac{k}{x}$.

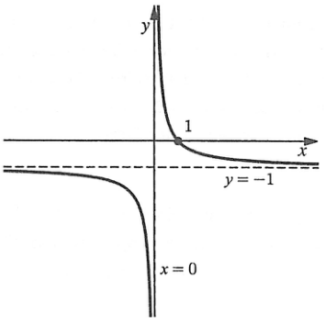


Question 2. Find the equation of each of the following graphs in the form $y = \frac{1}{x-a} + b$.

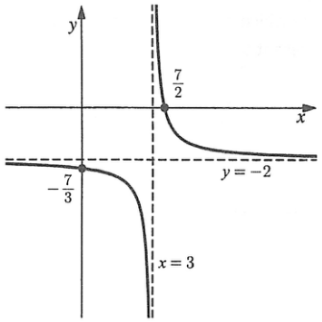
a)



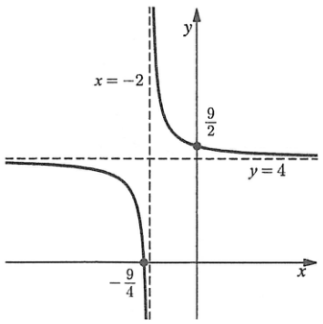
b)



c)



d)



Question 3. State the equations of the asymptotes of the curve $y = \frac{2}{x-1} + 3$.

Question 4. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

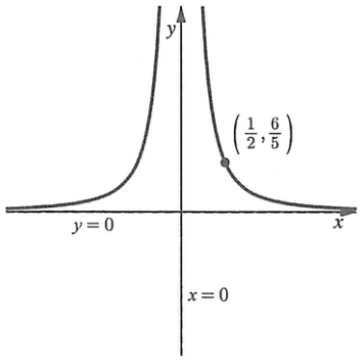
a) $y = \frac{1}{x} - 5$
b) $y = \frac{1}{x+3} - 1$
c) $y = \frac{4}{x-4} + 2$
d) $y = 3 - \frac{1}{x}$
e) $y = 5 - \frac{1}{x+3}$

Question 5. It is given that $k > 0$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \frac{k}{x} - 2k$

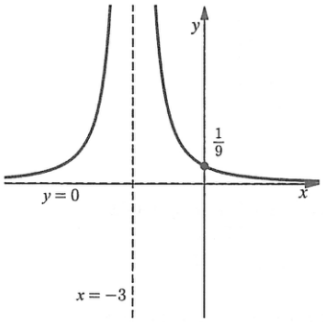
b) $y = k - \frac{1}{x-2k}$

Question 6. Find the equation of the following graph in the form $y = \frac{k}{x^2}$.

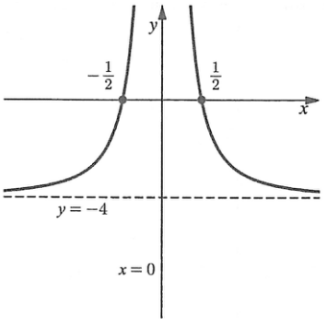


Question 7. Find the equation of each of the following graphs in the form $y = \frac{1}{(x-a)^2} + b$.

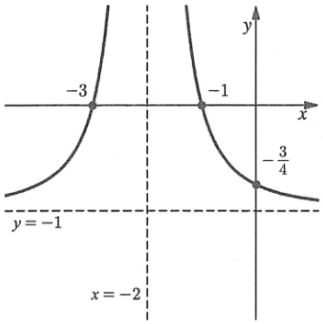
a)



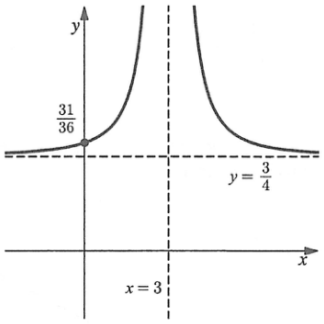
b)



c)



d)



Question 8. State the equations of the asymptotes of the curve $y = \frac{1}{(x+5)^2} - 2$.

Question 9. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \frac{1}{x^2} + 2$
b) $y = \frac{1}{(x-1)^2} - 4$
c) $y = \frac{4}{(x-4)^2} + 9$
d) $y = 4 - \frac{1}{x^2}$
e) $y = 9 - \frac{1}{(x+2)^2}$

Question 10. It is given that $k > 0$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \frac{1}{(x-k)^2} + 3k$
b) $y = 4k^2 - \frac{9}{(x+k)^2}$